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## DEPARTMENT OF CSE

**PROJECT ABSTRACT SUBMISSION**

**COURSE:24AD3107 DATA SCIENCE**

**A.Y. 2026 - 2027**

| **TITLE OF THE PROJECT** | **UPI Fraud Risk Scoring and Anomaly Detection** | |
|---|---|---|
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# Project Abstract

UPI Fraud Risk Scoring and Anomaly Detection is a data science and machine learning project that helps identify suspicious UPI transactions. As UPI is used for a large number of digital payments, checking every transaction manually is difficult. Our system analyzes transaction details such as amount, time, location, device, transaction frequency, and previous transaction history to find unusual activities.

The system uses Python, Pandas, and Scikit-learn for data processing and machine learning. We use MySQL to store user details, transaction records, and the results of fraud analysis. Machine learning algorithms such as Isolation Forest can be used to detect transactions that are different from normal user behavior.

For every transaction, the system generates a risk score between 0 and 100. Based on the score, transactions are classified as Low Risk, Medium Risk, or High Risk. For example, a high-value transaction from a new device at an unusual time may receive a higher risk score.

A Streamlit dashboard can be used to display transaction details, risk scores, suspicious transactions, and graphs. The dashboard helps users easily understand the transaction patterns and identify high-risk activities. It can also show the number of normal and suspicious transactions, risk levels, and changes in transaction behavior over time.

The main aim of the project is to detect unusual transaction behavior early and help identify potentially fraudulent transactions. This project combines data science, machine learning, anomaly detection, MySQL, and data visualization to solve a real-world digital payment security problem. It can help in making the monitoring process faster and provide useful information for further verification of suspicious transactions.